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[23CSE105]

FIRST SEMESTER B.Tech EXAMINATION, JANUARY-2025

DATA STRUCTURES USING C

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		Questions	Blooms Level	CO's	Marks
		PART-A	**5 x2=10**		
1.	**a.**	Define array.	L1	CO1	**02**
	b.	Classify the different types of data structures.	L1	CO4	**02**
	c.	List the different operations of the stack.	L2	CO3	**02**
	d.	List any 4 applications of queue.	L2	CO3	**02**
	e.	Define binary tree.	L1	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Write a program to find the factorial of n using recursion.	L3	CO2	**05**
		OR			
	b.	Write a program to find the Fibonacci series of n numbers using recursion.	L3	CO2	**05**
3.	**a.**	Explain the purpose of the functions malloc(), calloc(), realloc(), and free() in C.	L2	CO4	**05**
		OR			
	b.	Explain the types of linked list with its representation.	L2	CO4	**05**
4.	**a.**	Convert the expression (a+b)*c to postfix notation.	L3	CO3	**05**
		OR			
	b.	Write the code for the stack functions isfull() and isempty() using array.	L3	CO3	**05**
5.	**a.**	Explain the operations of queue with example.	L2	CO3	**05**
		OR			
	b.	Explain Double ended queue (DEQUE) with example.	L2	CO3	**05**
6.	**a.**	Traverse the given tree using Inorder, Preorder and Postorder traversals.	L3	CO5	**05**

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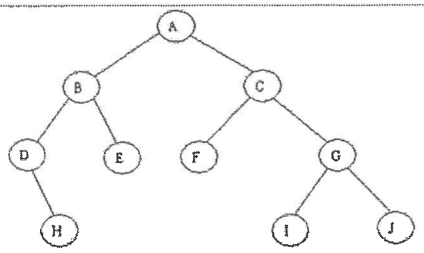
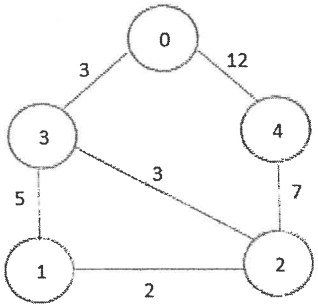
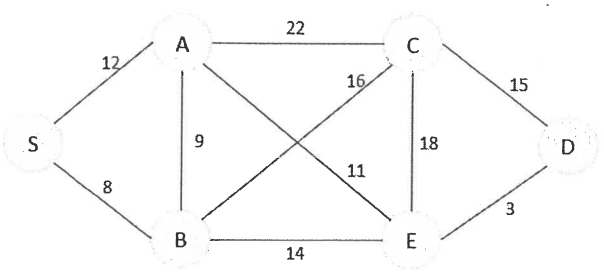
					
		OR			
	b.	Given the following weighted graph, use Kruskal's algorithm to find the MST. Show all steps involved in the process. 	L3	CO5	**05**
7.	**a.**	Explain extern storage class with an example.	L2	CO1	**05**
		OR			
	b.	Explain static storage class with an example.	L2	CO1	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Compare Call-by-value and Call-by-reference with example programs.	L3	CO1	**10**
		OR			
	b.	Compare the use of pointers with arrays in C. When would you prefer using pointers over arrays, and why?	L3	CO1	**10**
9.	**a.**	Write a C program to implement doubly linked list.	L3	CO4	**10**
		OR			
	b.	Write a C program to implement singly linked list.	L3	CO4	**10**
10.	**a.**	Write a C program to implement stack using array.	L3	CO3	**10**
		OR			
	b.	Write a C program to implement Queue using Array.	L3	CO3	**10**
11.	**a.**	Write a C program to implement stack using linked list.	L3	CO4	**10**
		OR			
	b.	Write a C program to implement Queue using Linked List.	L3	CO4	**10**

12.	**a.**	Construct the Binary Search Tree for the key elements – 45,15,79,90,10,55,12,20,50. Draw the tree structure after deleting a leaf node, parent node has only one child and parent node has two child nodes.	L3	CO5	**10**
		OR			
	b.	Find the minimum spanning tree using Prim's algorithm for the given graph.  	L3	CO5	**10**
13.	**a.**	Design a system that uses both stacks and queues to simulate a call center's customer support system, where calls are prioritized based on urgency (stack) and handled in the order they arrive (queue).	L4	CO6	**10**
		OR			
	b.	Design a custom data structure to efficiently manage a library system that supports operations like adding books, checking availability, and tracking borrowed books by users.	L4	CO6	**10**